

How Do You Say Your Name?

Sensitivity and Heterogeneity Analysis of Name Fluency and Academic Placement Outcomes

By Qiao Meng

I. Introduction

Ge and Wu (2024) study a relatively underexplored form of discrimination in the labor market, namely name fluency. More specifically, they examine the relationship between the difficulty of pronouncing a candidate's name and the probability of obtaining an academic job, the probability of securing a tenure-track position, and the research quality of the initial hiring institution. The central argument of the paper is that, in the labor market, name complexity itself may induce discriminatory responses from employers, thereby lowering a candidate's probability of being hired.

The first part of the paper uses data on economics PhD job market candidates in 2016 and 2017, which the authors manually collected from departmental websites. To measure pronunciation fluency, the authors construct three indicators: an algorithm-based score, the time required to pronounce the name, and subjective ratings of pronunciation difficulty. For employment outcomes, the first set of measures consists of two binary variables: whether the candidate obtains an academic position and whether the candidate secures a tenure-track position. The authors estimate probit models to examine the effect of pronunciation difficulty on these outcomes. They find that a one-standard-deviation increase in pronunciation difficulty reduces the probability of obtaining an academic job by 3 percentage points, with a p-value of 0.06. As a second measure of labor market outcomes, the authors estimate a tobit model using the RePEc ranking of the candidate's initial placement institution as the dependent variable, in order to test whether greater name difficulty is associated with worse placement quality. They find that a one-standard-deviation increase in pronunciation difficulty is associated with a decline of 67 places in the RePEc ranking of the hiring institution, and this result is statistically significant at the 5 percent level. The authors further note that this magnitude is roughly comparable to moving from the Princeton economics department to the economics department at UC Irvine.

The authors argue that the mechanism underlying these results is cognitive cost: as the difficulty of pronouncing a candidate's name increases, employers incur greater cognitive burden in processing the applicant, which in turn lowers the probability of being hired. They further provide supporting evidence for this mechanism through subgroup heterogeneity analyses. In particular, the paper shows that the name penalty

is concentrated among candidates with relatively weaker résumés, which is consistent with the cognitive-cost interpretation.

The second part of the paper reexamines data from two earlier audit studies and converts the names used in those experiments into algorithm-generated name fluency scores. In the Bertrand and Mullainathan (2004) data, the results provide additional support for the cognitive-cost hypothesis: the penalty associated with difficult-to-pronounce names is more severe among low-quality résumés. In the Oreopoulos (2011) data, the authors find that name difficulty can account for approximately 10 to 50 percent of the discrimination associated with immigrant names.

My analysis focuses on the first part of the paper, namely the section that uses job market data to study the effect of name difficulty on placement outcomes. Since the authors have already conducted extensive robustness checks, I instead focus on two additional analyses that are not covered in the original paper. The first is a sensitivity analysis based on assigning different weights to the first name and the last name. I find that when the full-name score is constructed using alternative weighting schemes, the estimated marginal effects do change systematically. However, the core conclusion remains robust in sign: regardless of whether the weighting scheme places greater emphasis on the first name or the last name, less fluent names are consistently associated with a lower probability of obtaining an academic placement.

The second is a heterogeneity analysis based on advisor quality. I find that the negative effect of name fluency tends to be stronger among candidates with lower-quality advisors, for whom external signals of applicant quality are weaker, and correspondingly weaker among candidates with higher-quality advisors. This pattern is also consistent with the cognitive-cost mechanism emphasized in the original paper: when the market lacks other strong signals of candidate quality, employers are more likely to be influenced by the negative impressions generated by name difficulty; by contrast, when candidates possess stronger external quality signals, this negative effect may be attenuated.

II. Data

The main dataset comes from information on economics PhD job market candidates manually collected by Ge and Wu. Specifically, the sample consists of 1,510 economics PhD candidates who entered the job market in the 2016–2017 and 2017–2018 cycles and who came from top-96 economics PhD programs. From candidates' CVs, the authors compile a range of personal and academic background information, including names, gender, graduate institution, fields of specialization, publication records, and prior work experience. In addition, through web searches, the authors collect information on each candidate's PhD advisor, including whether the advisor has served as an editor of a top-five journal, whether the advisor has published in top-five journals, and the number of top-five publications.

The labor market outcomes examined in the paper can be divided into three dimensions. First, whether the candidate obtains an academic job, which includes both tenure-track positions and other academic appointments such as postdoctoral positions or visiting appointments. Second, whether the candidate obtains a tenure-track position. Third, the research quality of the candidate's initial placement institution, measured by its RePEc ranking. In the tobit specification, the authors use this ranking as a proxy for placement quality. More specifically, the institutional ranking is used only when the candidate obtains a tenure-track position; for all other placements, the outcome is coded as 1,000. For institutions ranked in the top 6 percent, the score is set to 400; for the top 7 percent, 500; for the top 8 percent, 600; for the top 9 percent, 700; for the top 10 percent, 800; and for all remaining institutions, 1,000.

The paper also measures name fluency along three dimensions. The first is an algorithm-based measure derived from the procedure developed by Vitevitch and Luce (2004), which generates a pronunciation-difficulty score for each name. The authors calculate separate scores for first names and last names, and then simply add them together to construct their main full-name difficulty measure. The second measure is the median amount of time required for native English speakers to pronounce each name. The third is a subjective rating of pronunciation difficulty on a scale from 1 to 10 provided by independent raters.

The analysis that follows in this paper is centered on the authors' main result: when the algorithm-based full-name score is used as the measure of name difficulty, a one-standard-deviation increase in pronunciation difficulty reduces the probability of obtaining an academic job by 3 percentage points. Accordingly, the data used in my own analysis are also primarily focused on the dimensions corresponding to this main result, namely the algorithm-based measure of name pronunciation difficulty and the indicator for obtaining an academic job. The only modification is in my first sensitivity analysis: instead of directly using the full-name score constructed in the original paper by simply summing the first-name and last-name scores, I reconstruct the full-name score by assigning different weights to the first name and the last name.

III. Model Specification

For the binary employment outcomes in the original paper—namely, whether a candidate obtains an academic job and whether a candidate secures a tenure-track position—the authors employ a probit model. The model can be written as:

$$Pr(Placement_i = 1) = \Phi(\delta' Pronounce_i + \phi' X_i + \theta' Z_i)$$

where $Pronounce_i$ denotes the measure of name pronunciation difficulty. In the original paper, the authors use three alternative measures: the algorithm-based score,

pronunciation time, and subjective ratings. Different specifications use different measures depending on the empirical exercise.

X_i represents the candidate's individual characteristics, including gender, whether the candidate attended an elite undergraduate institution in the United States, the total number of publications, the number of top-five publications, the number of revise-and-resubmit (R&R) papers, the number of top-five R&Rs, whether the candidate received a teaching award, and whether the candidate has independent teaching experience.

Z_i captures characteristics of the candidate's PhD advisor, including the advisor's number of top-five journal publications and whether the advisor has served as a journal editor, especially whether the advisor has served as an editor of a top-five journal.

In addition, the model controls for a range of fixed effects and covariates that may affect PhD placement outcomes, including undergraduate region, specific field of specialization (such as theory, macroeconomics, finance, econometrics, and applied microeconomics), job market cycle (2016 or 2017), and PhD-granting institution.

To analyze placement quality—that is, the non-binary employment outcome measured by the RePEc ranking of the initial hiring institution—the authors use a tobit model. Because the ranking variable is upper-censored by construction, the authors set the upper bound of the placement outcome at 1,000 and estimate a right-censored tobit model. Similar to the probit specification above, the tobit model also includes controls for candidate characteristics, advisor characteristics, name fluency measures, and fixed effects.

IV. Analysis Design

A. Sensitivity Analysis

This section examines the following question: when the full-name score is constructed using different weighting schemes, do the estimated marginal effects change systematically?

Holding the baseline regression specification fixed, I vary only the construction of the full-name difficulty index in order to assess whether the main result depends on the specific definition of this key explanatory variable. In the original paper, the difficulty of a full name is defined as the simple sum of first-name difficulty and last-name difficulty, which implicitly assumes that the two components contribute equally to overall name difficulty. Although this approach is intuitive, it also imposes a fairly

strong restriction—namely, that employers or evaluators place symmetric weights on the first name and the last name when processing a candidate’s name.

However, both actual communication patterns and common intuition suggest that first names and last names may not play identical roles in name recognition, pronunciation, and memory. I therefore relax this equal-weight assumption and allow the first name and the last name to enter the composite full-name score with different weights. Specifically, I redefine the full-name score as:

$$FullNameScore_i(w) = w \cdot FirstNameScore_i + (1 - w) \cdot LastNameScore_i$$

where $FirstNameScore_i$ denotes the algorithm-based fluency score of individual i ’s first name, $LastNameScore_i$ denotes the corresponding score for the last name, w is the weight placed on the first name, and $1 - w$ is the weight placed on the last name.

In the implementation, I allow w to vary from 0 to 1 in increments of 0.01, generating 101 alternative weighting schemes. Under each scheme, I re-estimate the marginal effect of the full-name score on the probability of academic placement. This design allows me to assess systematically whether the main conclusion of the original paper is robust across a range of plausible index constructions, or whether it is sensitive to the specific weighting rule used to construct the full-name index.

To further examine whether the estimated effects vary systematically with the modeling choice w , I conduct an internal meta-regression using the 101 estimated marginal effects.

First, I estimate the following linear meta-regression model:

$$\hat{\beta}(w) = a + bw + u_w$$

where $\hat{\beta}(w)$ is the estimated marginal effect obtained under weighting scheme w , and u_w is the error term. This specification is used to test whether the estimated effect changes linearly with the weight placed on the first name.

Because the graphical evidence suggests that the relationship may be nonlinear, I further estimate the following quadratic specification:

$$\hat{\beta}(w) = a + b_1w + b_2w^2 + u_w$$

This specification allows for a nonlinear relationship between the marginal effect and the weighting parameter, and therefore provides a more complete test of whether the estimated effects vary systematically across alternative weighting schemes.

B. Heterogeneity Analysis

The starting point of the heterogeneity analysis is one of the paper's central mechanisms: the penalty associated with name fluency is partly driven by cognitive cost. When employers face a difficult-to-pronounce name, they may be more likely to rely on the negative impression associated with low fluency when other strong signals of candidate quality are absent.

By contrast, an advisor with a strong academic reputation can be viewed as a credible quality signal, which may partially offset the negative effect associated with a difficult-to-pronounce name.

Motivated by this idea, I conduct the heterogeneity analysis by splitting the sample according to advisor quality. Among the available measures of advisor quality, I focus on two indicators: whether the advisor has any top-five publications and whether the advisor has served as an editor of a top-five journal. I then estimate the model separately by subgroup and compare the sign, magnitude, and statistical significance of the estimated name penalty across groups.

V. Analysis Results

A. Sensitivity Analysis

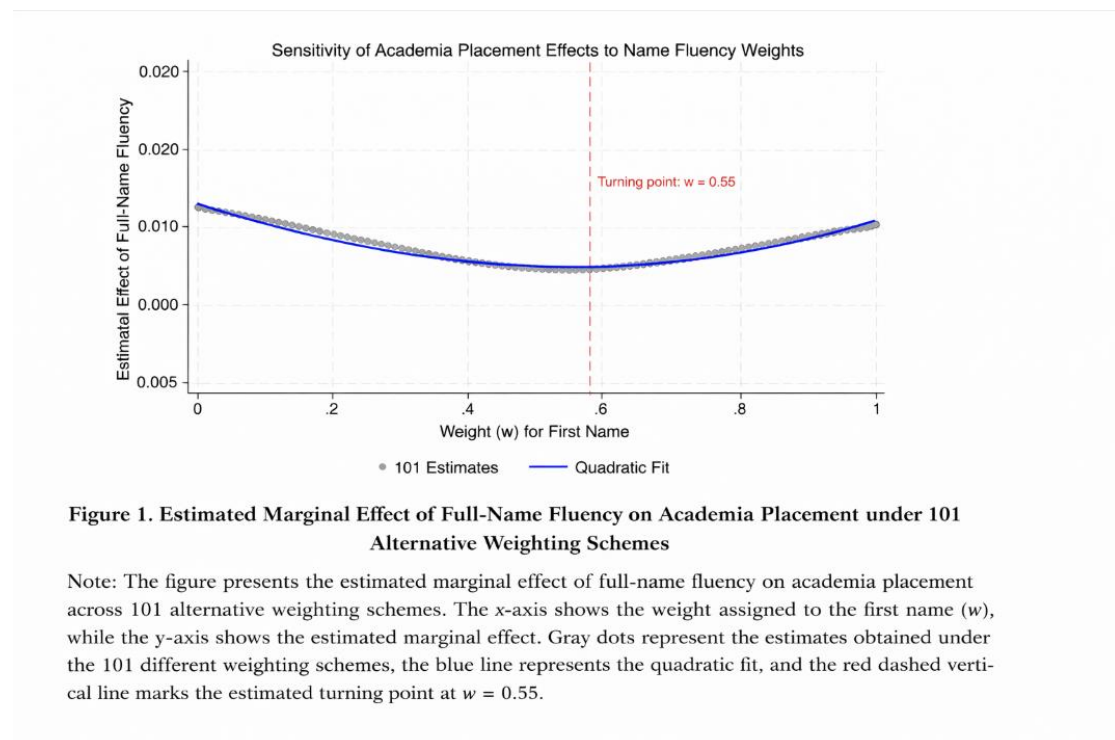


Figure 1 plots the relationship between the estimated marginal effects under 101 alternative weighting schemes and the weight placed on the first name. The horizontal axis shows the first-name weight w , while the vertical axis shows the estimated marginal effect of the full-name score on academic placement. The gray dots correspond to the estimated effects under the 101 alternative weighting schemes, the blue line represents the quadratic fitted curve, and the red dashed line indicates the estimated turning point.

Figure 1 reveals two clear patterns.

First, all estimated effects are below zero, indicating that the main conclusion is robust in sign. Regardless of whether the full-name score places greater weight on the first name or the last name, more difficult-to-pronounce names are consistently associated with a lower probability of obtaining an academic placement. This suggests that changing the construction of the full-name score does not overturn the baseline result.

Second, the variation is not monotonic; instead, it exhibits a pronounced nonlinear pattern. As the weight on the first name increases from zero, the estimated name penalty becomes progressively more negative, implying that the adverse effect strengthens. Once the weight enters the middle range, the negative effect reaches its maximum magnitude. As the weight continues to rise, so that the full-name score is driven more heavily by first-name difficulty, the negative effect begins to weaken.

Table 1. Internal Meta-Regression of Marginal Effects on Alternative Name Weighting Schemes

Internal Meta-Analysis of Name Fluency Weights and Academia Placement Effects		
	(1)	(2)
VARIABLES	Linear Model Quadratic Model	Linear Model Quadratic Model
w	-0.004*** (0.001)	-0.046*** (0.001)
w_sq		0.042*** (0.001)
Constant	-0.023*** (0.001)	-0.016*** (0.000)
Observations	101	101
R-squared	0.125	0.978
Dependent Variable	Marginal Effect	Marginal Effect
Standard errors in parentheses		
*** p<0.01, ** p<0.05, * p<0.10		

Notes: The table reports internal meta-regression results based on 101 alternative constructions of the full-name fluency score. Column (1) reports the linear specification, and column (2) reports the quadratic specification. The dependent variable is the estimated marginal effect of the full-name score on academic placement. Standard errors are reported in parentheses. The implied turning point from the quadratic model is approximately $w = 0.55$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

This graphical pattern is further confirmed by the internal meta-regression results reported in Table 1. The linear meta-regression explains only a relatively small share of the across-specification variation, with an R^2 of only 0.125. This suggests that a purely linear characterization of the relationship between the weighting parameter and the estimated marginal effect misses most of the systematic variation. By contrast, the quadratic specification provides an almost complete fit to the sequence of estimates, with an R^2 of 0.978. This indicates that the variation in marginal effects across alternative weighting schemes is not random noise, but instead follows a highly structured nonlinear pattern.

The coefficients from the quadratic specification further show that the coefficient on w is negative, while the coefficient on w^2 is positive, implying a clear U-shaped relationship between the first-name weight and the estimated marginal effect. Based on the estimated quadratic coefficients, the implied turning point is approximately $w = 0.55$. This means that the estimated name penalty is strongest when the weight on the first name is slightly greater than one-half. In other words, within the setting of this study, the full-name score does not capture the name-fluency effect most strongly when it places extreme weight on either the first name or the last name; rather, the strongest negative association with academic placement arises when the first name receives slightly more weight than the last name.

Overall, this analysis clearly shows that when the full-name score is constructed using different weighting schemes, the estimated marginal effect does vary systematically. At the same time, however, this variation mainly takes the form of a nonlinear adjustment in the magnitude of the negative effect, rather than a change in the direction of the result. In other words, alternative weighting schemes affect the size of the estimated effect, but they do not alter the core finding that name difficulty is negatively associated with academic placement.

B. Heterogeneity Analysis

Table 2: NAME FLUENCY AND PLACEMENT OUTCOMES BY TOP 5 EDITOR STATUS OF ADVISORS

	Advisor is Top 5 Editor			Advisor is not Top 5 Editor		
	Academia (1)	TT (2)	RePEc (3)	Academia (4)	TT (5)	RePEc (6)
combined_total	-0.014 (0.048)	-0.034 (0.052)	155.964** (70.621)	-0.035** (0.018)	-0.007 (0.018)	45.598 (36.364)
Observations	171	175	188	1,281	1,311	1,322
Control for Name Length	Yes	Yes	Yes	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Subfield/Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Region/JM Cycle FE	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 3: NAME FLUENCY AND PLACEMENT OUTCOMES BY TOP 5 PUBLICATION OF ADVISORS

	No Top 5 publication			Has Top 5 publication		
	Academia (1)	TT (2)	RePEc (3)	Academia (4)	TT (5)	RePEc (6)
combined_total	-0.111** (0.044)	-0.082** (0.041)	0.924 (119.933)	-0.024 (0.019)	0.001 (0.020)	74.585** (32.567)
Observations	264	298	350	1,113	1,126	1,160
Control for Name Length	Yes	Yes	Yes	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes	Yes	Yes	Yes
Subfield/Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Region/JM Cycle FE	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The results reported in Tables 2 and 3 generally support this conjecture. First, when the sample is split according to whether the advisor has any top-five publications, a clear pattern emerges. In the No Top 5 Publication group, the coefficient on combined total is -0.111 for academia and -0.082 for tenure-track placement, both statistically significant at the 5 percent level. By contrast, in the Has Top 5 Publication group, the coefficient declines to -0.024 for academia, while the coefficient for tenure-track placement is essentially zero, at 0.001; neither estimate is statistically significant. At the same time, the coefficient for the RePEc outcome is positive and statistically significant in the Has Top 5 Publication group, but insignificant in the group without top-five publications.

Second, a broadly similar pattern appears when the sample is split according to whether the advisor has served as an editor of a top-five journal. In the Top 5 Editor Advisor group, the coefficient on combined total is -0.014 for academia and -0.034 for tenure-track placement, and neither estimate is statistically significant. In the Non-Top 5 Editor Advisor group, the coefficient for academia is -0.035, which is

statistically significant at the 5 percent level, while the coefficient for tenure-track placement is -0.007 and statistically insignificant. For the RePEc outcome, the coefficient is positive and statistically significant in the Top 5 Editor Advisor group, but insignificant in the non-top-5-editor group.

In terms of coefficient signs and relative magnitudes, these results suggest that advisor quality does indeed affect how the name fluency penalty manifests itself. In particular, when advisor quality is measured using top-five publication status, the penalties for academia and tenure-track placement are clearly stronger in the lower-quality-advisor group, while the negative effects are substantially attenuated in the higher-quality-advisor group. When top-five editor status is used as the grouping criterion, the pattern is less clean than in the publication split, but it still suggests that the effect of name difficulty is not uniformly distributed across candidates; rather, it varies with the strength of advisor-based quality signals.

Overall, this heterogeneity analysis indicates that the negative effect of name fluency tends to be stronger among candidates with weaker advisors, for whom external signals of quality are more limited, and relatively weaker among candidates with stronger advisors. These findings are consistent with the cognitive-cost mechanism emphasized in the original paper: the observed name penalty may partly operate through the additional mental effort required for employers to process and remember difficult-to-pronounce names, and employers may be more willing to incur this cost for candidates who possess stronger external quality signals. In the context of this paper, advisor quality can be interpreted as one such external signal of candidate quality.

That said, it should be noted that the current evidence is based primarily on subgroup regressions. As a result, the analysis is informative about patterns of heterogeneity, but it does not by itself establish that the differences across groups are statistically significant.

VI. Conclusion

In this paper, I revisit the first part of Ge and Wu (2024) and focus on the effect of name fluency on academic placement outcomes for economics PhD candidates. My sensitivity analysis shows that the estimated marginal effect changes systematically when the full-name score is constructed using alternative weights for first and last names, but the main conclusion remains unchanged: less fluent names are consistently associated with a lower probability of academic placement. This pattern is well summarized by an internal meta-regression, which indicates a clear nonlinear relationship between the weighting scheme and the estimated effect. I also find

evidence of heterogeneity by advisor quality. The negative effect of name fluency tends to be stronger among candidates with weaker advisor signals and smaller among those with stronger advisor signals, which is consistent with the cognitive-cost mechanism emphasized in the original paper. Overall, my results support the authors' main argument while showing that both the construction of the fluency measure and the strength of external quality signals matter for the magnitude of the estimated penalty.

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